

A* Examination Guide

Built from 18 past papers and their mark schemes, October 2020 to June 2026

This guide is built from the P4 question papers and mark schemes themselves. Methods are set out the way the mark schemes award M and A marks, and every warning comes from a note that examiners applied: “A0”, “withhold”, “do not allow”, “answers with no working score no marks”. Use the formula booklet for what it gives you, and learn the rest.

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|---|--|----|--|
| 1 | How P4 is built | 7 | Parametric equations |
| 2 | Twenty mark-scheme rules for P4 | 8 | Integration techniques |
| 3 | Proof by contradiction | 9 | Areas and volumes of revolution |
| 4 | Partial fractions | 10 | Differential equations and connected rates |
| 5 | Binomial expansion | 11 | Vectors |
| 6 | Implicit and logarithmic differentiation | 12 | Final A* checklist |

Papers analysed: Oct 2020, Jan 2021, Jun 2021, Oct 2021, Jan 2022, Jun 2022, Oct 2022, Jan 2023, Jun 2023, Oct 2023, Jan 2024, Jun 2024, Oct 2024, Jan 2025, Jun 2025, Oct 2025, Jan 2026, Jun 2026. Papers are cited by series, e.g. (Jun 24).

1 HOW P4 IS BUILT

Format: 1 hour 30 minutes, 75 marks, 8–11 questions: about 1.2 minutes per mark. Many parts are marked “Solutions relying on calculator technology are not acceptable”, so exact algebraic working is the norm. P4 builds on P3: chain, product and quotient rules and standard integrals are assumed.

Topic	Papers (of 18)	What is asked most
Binomial expansion (n not a positive integer)	18	first 3–4 terms, validity, approximations, unknown k or n
Vectors in 3D	18	intersection/skew, angle, foot of perpendicular, area, reflection
Integration by substitution and by parts	18	exact definite integrals, often “p + q ln 2”
Parametric equations	18	dy/dx, tangent/normal, Cartesian form, range, area
Implicit differentiation	17	dy/dx, tangent/normal, turning points, max x or y
Differential equations	17	separate variables, partial fractions, model in context
Volumes of revolution	17	$\pi y^2 dx$, often parametric or after division
Proof by contradiction	15	parity, irrational roots, no intersection, no stationary points
Partial fractions	14	repeated factor, improper fractions, with integration or binomial
Connected rates of change	12	spheres, cones, cubes, bowls: $dV/dt = dV/dh \times dh/dt$

The recurring long questions

- **Vectors (8–12 marks):** every paper. Typical chain: equation of a line → intersection or skew → angle → nearest point → area or reflection.
- **Differential equation in context (7–11):** tank depth (Jan 21), ice ball (Oct 22), engine temperature (Jun 23), goats (Oct 23), circuit current (Jun 24), fairground ride (Oct 24), balloon (Jan 26), algae (Jun 26).
- **Parametric curve with area or volume (8–11):** the last question on most papers.
- **Proof by contradiction (4–6):** often a student's partial proof to complete (Oct 21, Oct 22, Jan 23, Oct 25).

2 TWENTY MARK-SCHEME RULES FOR P4

1. **Calculator-only answers score 0** where the question bans calculator technology: “16 with no working scores no marks. This is evidence of use of a calculator” (Oct 20).
2. **Proof by contradiction needs a written assumption** (“Assume ...”, “Let ...” in words), the algebra, a clear statement of the contradiction, and a conclusion. “Contradiction” alone is not enough.
3. **Skew lines need two facts:** the lines do not intersect (inconsistent third equation shown) **and** they are not parallel (direction vectors not multiples). Saying “skew” early does not score.
4. **Angles between vectors:** use the scalar product with both vectors leaving the same point. Radians score A0 when degrees are asked (Jan 21). For the angle between lines use the direction vectors and take the acute angle.
5. **Vector notation:** do not mix i, j, k with column vectors, or write coordinates where a vector is asked (“do not accept coordinates”, Oct 22). Hybrid forms such as $5i, 6j, -2k$ lose the mark (Jun 24).
6. **Binomial coefficients in simplest form;** “+–” written for “–” scores A0 (Jan 21, Jan 22).
7. **Validity:** state $|x| < \dots$ for every expansion; for a product or sum of expansions use the narrower interval.
8. **Approximations must use the expansion:** substitute the given x into your series and show it.
9. **Implicit differentiation:** every term differentiated, constants → 0, and the product rule on xy terms. The “= 0” in a final line equation must be seen.
10. **Turning point:** set $dy/dx = 0$ (numerator = 0) and use it with the curve equation. Maximum x on a curve comes from $dx/dy = 0$, i.e. the denominator of $dy/dx = 0$ (Jun 26).
11. **Parametric gradient** = $(dy/dt) \div (dx/dt)$, found at the value of t for the point.
12. **Substitution must change everything:** the integrand, $dx \rightarrow du$, and the limits. “ $u = x$ is not acceptable” as a substitution (Oct 23).
13. **Limits in parametric areas and volumes** are values of t : convert x -limits to t -limits first.
14. **Exact answers:** keep π , $\ln$ and surds. Upper limit $2 \ln 2$ or $\ln 4$, “do not accept 1.386” (Oct 20).
15. **+ c** on every indefinite integral, including in differential equations before using the condition.
16. **$\ln|\dots|$ with modulus** where the argument can be negative; condone brackets only in some schemes.
17. **Differential equations:** separate correctly (dy and dx in the right places), integrate both sides, find c from the condition. Substituting the condition into a given answer scores nothing (Jan 22).
18. **Connected rates:** substitute the quantity asked for; “substituting $h = 1.5$ or $h = 90$ will score M0M0” when $t = 90$ s was given (Oct 24).
19. **Reject impossible values** with a reason ($k = 0$ when $k \neq 0$ was stated; the root not matching the diagram).
20. **Given answers (“show that”)** need every step and no errors, including brackets.

3 PROOF BY CONTRADICTION

STRUCTURE EXAMINERS MARK (4–6)

1. Assume the opposite, in words: “Assume there exists an integer n such that n^3 is even and n is odd.” **2. Use** the assumption algebraically. **3. Contradict:** state what the algebra shows and why it is impossible. **4. Conclude:** “so the assumption is false, hence n^3 even $\Rightarrow n$ even.”

MODEL ANSWER · N^3 EVEN $\Rightarrow N$ EVEN (OCT 20, OCT 21)

Assume n^3 is even and n is odd, so $n = 2p + 1$ for an integer p . Then $n^3 = 8p^3 + 12p^2 + 6p + 1 = 2(4p^3 + 6p^2 + 3p) + 1$, which is odd. This contradicts n^3 being even, so if n^3 is even then n is even.

MODEL ANSWER · $\sqrt[3]{2}$ IS IRRATIONAL (OCT 21; $\sqrt[3]{3}$ IN JAN 23)

Assume $\sqrt[3]{2} = a/b$ where a and b are integers with no common factor. Then $a^3 = 2b^3$, so a^3 is even, so a is even (previous part): $a = 2k$. Then $8k^3 = 2b^3 \Rightarrow b^3 = 4k^3$, so b^3 and hence b are even. a and b share the factor 2: contradiction. So $\sqrt[3]{2}$ is irrational.

- **Curves do not meet** (Oct 24, Jun 24): assume they intersect, equate, rearrange to something impossible, e.g. $x^4 + 8x^2 + 15 = 0$, impossible since $x^4, x^2 \geq 0$. Refer to both terms being non-negative.
- **No stationary points** (Jan 24): assume $dy/dx = 0$ has a solution, e.g. $2 + 3x^2 - \sin x = 0 \Rightarrow \sin x = 2 + 3x^2 \geq 2$, impossible as $|\sin x| \leq 1$.
- **Sequence not geometric** (Jan 22): assume a common ratio, form $(1 + 2k)^2 = k(3 + 3k)$, show the quadratic has no real roots (discriminant < 0).
- **Completing a student's proof** (Oct 22, Jan 23): finish every case (e.g. $p = 3k + 2$ as well as $3k + 1$) and state the contradiction and conclusion.
- A counter-example disproves a statement (Oct 23: the inequality fails for negative k).

4 PARTIAL FRACTIONS

Denominator	Form
$(ax + b)(cx + d)$	$A/(ax + b) + B/(cx + d)$
$(ax + b)(cx + d)^2$	$A/(ax + b) + B/(cx + d) + C/(cx + d)^2$
degree of numerator $\geq$ degree of denominator	divide first: polynomial + proper fraction

WORKED EXAMPLE · IMPROPER FRACTION THEN INTEGRATE (OCT 20)

$$(6x^2 - 16)/((x + 1)(2x - 3)) = 3 + 2/(x + 1) - 1/(2x - 3).$$

$\int \dots dx = 3x + 2 \ln|x + 1| - \frac{1}{2} \ln|2x - 3| + c$. The $+ c$ was required.

- Find constants by substituting the roots of each factor (cover-up), then compare coefficients for any remaining constant (e.g. the x^2 coefficient for a repeated factor).
- Jan 23: $(5x + 10)/((1 - x)(2 + 3x)) = 3/(1 - x) + 4/(2 + 3x)$; the expansion is valid for $|x| < 2/3$, the narrower of $|x| < 1$ and $|x| < 2/3$.

5 BINOMIAL EXPANSION

THE EXPANSION (FORMULA BOOKLET)

$$(1 + x)^n = 1 + nx + n(n - 1)x^2/2! + n(n - 1)(n - 2)x^3/3! + \dots, \text{ valid for } |x| < 1.$$

For $(a + bx)^n$: take out a^n first: $(a + bx)^n = a^n(1 + bx/a)^n$, valid for $|x| < |a/b|$. Put the whole of (bx/a) in brackets before squaring or cubing.

WORKED EXAMPLES

Jan 24: $(1 - 4x)^{-3} = 1 + 12x + 96x^2 + 640x^3 + \dots$, valid for $|x| < \frac{1}{4}$.

Oct 25: $(2 + 5x)^{-2} = \frac{1}{4}(1 + 5x/2)^{-2} = \frac{1}{4} - (5/4)x + (75/16)x^2 - (125/8)x^3 + \dots$, valid for $|x| < 2/5$.

- Unknown n or k (Jun 21, Jun 22, Jan 26): match the given coefficients, e.g. nk = coefficient of x , and solve; reject $k = 0$ if it is excluded.
- Approximations (Jan 21, Jun 21, Oct 22, Oct 24): choose x inside the interval of validity so that the bracket gives the required root; show the substitution into the series.
- With partial fractions (Jan 23, Jun 26): expand each fraction and add.
- Products such as $(4 + 5x)^{1/2}(4 - 5x)^{1/2}$ (Jun 25): multiply the two series and keep terms up to the power asked.

6 IMPLICIT AND LOGARITHMIC DIFFERENTIATION

METHOD

$d/dx[f(y)] = f'(y) dy/dx$; $d/dx(xy) = y + x dy/dx$; $d/dx(y^2) = 2y dy/dx$; $d/dx(a^y) = a^y \ln a dy/dx$. Collect the dy/dx terms on one side and factorise.

WORKED EXAMPLE · TANGENT (JAN 22)

$xy^2 = x^2y + 6$ at $P(2, 3)$: $y^2 + 2xy y' = 2xy + x^2 y' \Rightarrow y' = (2xy - y^2)/(2xy - x^2) = 3/8$.

Tangent: $y - 3 = (3/8)(x - 2) \Rightarrow 3x - 8y + 18 = 0$.

- Turning points (Jun 22, Jan 24): numerator of $dy/dx = 0$ gives a relation between x and y ; substitute into the curve equation.
- Points furthest north/south (Oct 22) are where $dy/dx = 0$; maximum x (Jun 26) is where $dx/dy = 0$.
- Logarithmic differentiation (Oct 20): $y = x^{\sin x} \Rightarrow \ln y = \sin x \ln x \Rightarrow (1/y) dy/dx = \cos x \ln x + (\sin x)/x$. Stationary points: $\tan x + x \ln x = 0$.
- Normals: gradient $-1/m$; give integers with “= 0” when asked.

7 PARAMETRIC EQUATIONS

TOOLKIT

Gradient: $dy/dx = (dy/dt) \div (dx/dt)$. Find t at the point first.

Cartesian form: eliminate t , e.g. make t the subject, or use $\sin^2 t + \cos^2 t = 1$, $\cos 2t = 1 - 2\sin^2 t$, $\sec^2 t = 1 + \tan^2 t$.

Domain and range of the Cartesian form: from the range of x and y over the given t .

Area: $\int y dx = \int y (dx/dt) dt$ with t -limits. **Volume:** $\pi \int y^2 (dx/dt) dt$.

WORKED EXAMPLE · OCT 20 Q4

$x = 2t^2 - 6t$, $y = t^3 - 4t$. $y = 0$ at $t = 0, \pm 2$: $A(-4, 0)$ from $t = 2$, $B(20, 0)$ from $t = -2$.

$dy/dx = (3t^2 - 4)/(4t - 6) = -4/7$ at $t = -2$. Tangent: $7y + 4x - 80 = 0$.

Meets the curve again: $7(t^3 - 4t) + 4(2t^2 - 6t) - 80 = 0 \Rightarrow (t + 2)^2(7t - 20) = 0 \Rightarrow t = 20/7$, $x = -40/49$. The tangent point gives the repeated root.

- Rational parametric forms (Jan 21, Oct 25, Jun 26): write $y = (ax + b)/(cx + d)$ by making t the subject in x and substituting; the range follows from the t -domain.

- “Show that the area of $R = \int \dots$ ” (Jun 21, Oct 24, Jan 26): show $y \times dx/dt$, the simplification, and the new limits.

8 INTEGRATION TECHNIQUES

INTEGRATION BY PARTS

$\int u (dv/dx) dx = uv - \int v (du/dx) dx$. Choose $u = \ln x$ if present; otherwise $u =$ the polynomial.

Oct 20: $\int \ln x/x^2 dx = -(\ln x)/x - 1/x + c$. **Jun 24:** $\int x \cos 3x dx = (x/3) \sin 3x + (1/9) \cos 3x + c$.

$x^2 e^x$ or $x^2 e^{-x}$ (Jun 22, Jun 23, Jan 25): apply parts twice. $e^x \cos 2x$ (Oct 22, Jun 26): apply twice, then solve for the original integral.

INTEGRATION BY SUBSTITUTION

Write x and dx in terms of u , change the limits, simplify, integrate, and do not substitute back for a definite integral.

Typical: $u = 2x - 1$ in $\int 3x/\sqrt{2x - 1} dx$: $x = (u + 1)/2$, $dx = \frac{1}{2} du$, integrand $\rightarrow (3/4)(u^{\frac{1}{2}} + u^{-\frac{1}{2}})$.

Trig substitutions: $x = 2 \sin u$ (Jun 22), $x = 4 \sin \theta$ (Jan 25), $2x = \sin u$ (Jun 26); use $\cos^2 = 1 - \sin^2$.

Exponential: $u = e^x - 3$ (Oct 22), $u = e^x + 4$ (Jan 26): $dx = du/(u \pm k)$.

- Results to recognise: $\int f'(x)/f(x) dx = \ln|f(x)|$; $\int f'(x)[f(x)]^n dx = [f(x)]^{n+1}/(n + 1)$.
- Trig powers: use $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$, $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$, $\sin x \cos x = \frac{1}{2} \sin 2x$.
- Final forms such as $p + q \ln 2$ or $\ln(a/b)$: combine logs with the laws and simplify fractions fully.

9 AREAS AND VOLUMES OF REVOLUTION

FORMULAE

Area under $y = f(x)$: $\int y dx$. Volume about the x -axis: $V = \pi \int y^2 dx$ (parametric: $\pi \int y^2 (dx/dt) dt$).

For a region between a curve and a line, find the limits from the intersection, then add or subtract a cylinder, cone or triangle as needed.

WORKED EXAMPLE · OCT 20 Q3

$y = e^{0.5x} - 2$, rotated about the x -axis between $x = 0$ and $x = 2 \ln 2$ (where $y = 0$).

$V = \pi \int (e^{0.5x} - 2)^2 dx = \pi [e^{0.5x} - 8e^{0.5x} + 4x]$ from 0 to $2 \ln 2 = \pi(8 \ln 2 - 5)$.

The upper limit must be given as $2 \ln 2$ or $\ln 4$, not 1.386.

- Square y correctly before integrating: $(e^{0.5x} - 2)^2$ has three terms.
- Context volumes (Jan 22 doorknob, Jun 22 weight, Jun 25 vase): finish with the units and the accuracy asked; density = mass $\div$ volume.
- When y^2 is a fraction, divide or use partial fractions before integrating (Oct 24, Jun 25).

10 DIFFERENTIAL EQUATIONS AND CONNECTED RATES

SEPARATING VARIABLES

$dy/dx = f(x)g(y) \Rightarrow \int 1/g(y) dy = \int f(x) dx$. Add $+ c$ on one side, use the condition straight away, then rearrange to the form asked ($y = \dots$, $y^2 = \dots$).

Proportionality: “rate of decrease is inversely proportional to r^2 ” $\Rightarrow dr/dt = -k/r^2$ (Oct 22).

WORKED EXAMPLE · JAN 21 Q10 (TANK)

$dH/dt = -(H - 5)(H + 3)/40$, $H(0) = 13$. Partial fractions: $1/((H - 5)(H + 3)) = (1/8)[1/(H - 5) - 1/(H + 3)]$.

Integrating: $\ln|(H - 5)/(H + 3)| = -0.2t + c \Rightarrow (H - 5)/(H + 3) = Ae^{-0.2t}$; $t = 0$ gives $A = 1/2$. Rearranging: $H = (10 + 3e^{-0.2t})/(2 - e^{-0.2t})$.

$H = 8 \Rightarrow e^{-0.2t} = 6/11 \Rightarrow t = 5 \ln(11/6) = 3.03$ days. Long term: $e^{-0.2t} \rightarrow 0$, so $H \rightarrow 5$ m.

CONNECTED RATES · OCT 24 (CONE)

$r/h = 12/30 \Rightarrow r = 2h/5$, so $V = (1/3)\pi r^2 h = 4\pi h^3/75$. With $dV/dt = 2\pi$, after 90 s $V = 180\pi \Rightarrow h = 15$.

$dh/dt = (dV/dt) \div (dV/dh) = 2\pi \div (4\pi h^2/25) = 1/18 \text{ cm s}^{-1}$ at $h = 15$. Using $h = 90$ scored M0M0.

Balloon (Jan 25): $dV/dt = 70\pi$, $dV/dr = 4\pi r^2 = 100\pi$ at $r = 5$, so $dr/dt = 0.7 \text{ cm s}^{-1}$.

- Long-term value: let $t \rightarrow \infty$ in the solution (goats, algae, current). Interpret in context.
- “Will it ever be full?” (Oct 25): compare the limiting value with the capacity and give the reason.
- Forming the DE from a tank (Oct 21): rate in – rate out = $A dh/dt$, with A the cross-section area.

11 VECTORS**TOOLKIT**

Line: $r = a + \lambda d$. **Unit vector:** $d/|d|$. **Distance:** $|AB| = \sqrt{x^2 + y^2 + z^2}$.

Angle: $\cos \theta = (a \cdot b)/(|a||b|)$; take vectors pointing away from the vertex. Acute angle between lines: use $|a \cdot b|$.

Perpendicular: $a \cdot b = 0$. **Foot of perpendicular from P to a line:** write a general point $X(\lambda)$, set $PX \cdot d = 0$, solve for λ .

Intersection: equate components, solve two equations, check the third. If it fails, the lines do not meet.

MODEL ANSWER · PROVE SKEW (JAN 21, OCT 22, JUN 26)

Equate components: solve two of the equations for λ and μ , substitute into the third, and show it is not satisfied, so the lines do not intersect. The direction vectors are not multiples of each other, so the lines are not parallel. Hence the lines are skew.

WORKED EXAMPLES

Jan 21: $AB = 6i - 2j + 3k$, $BC = 2i + 5j + 8k$. Angle ABC uses $BA \cdot BC = -26$: $\cos B = -26/(7\sqrt{93})$, $B = 112.65^\circ$. Area of parallelogram = $|AB \times BC| = 62.3$.

Jun 21: $l: r = (1, -10, -9) + \lambda(4, 4, 2)$. Nearest point to O: $(1 + 4\lambda, -10 + 4\lambda, -9 + 2\lambda) \cdot (4, 4, 2) = 0 \Rightarrow \lambda = 1.5$, $X(7, -4, -6)$. Since $OX \perp l$ and OA is a unit vector along l , area $OXA = \frac{1}{2} \times \sqrt{101} \times 1 = \frac{1}{2}\sqrt{101}$.

- Reflection of A in a line (Oct 21, Jun 23): find the foot X, then $OB = 2OX - OA$.
- Shortest distance from A to $l = |AX|$ with X the foot of the perpendicular.
- Points dividing a line in a ratio (Jun 21): $AB : AC = 1 : 3 \Rightarrow AC = 3AB \Rightarrow c = 3b - 2a$.
- Two possible points at a given distance (Jun 24, Jan 25): form a quadratic in λ and give both points.

12 FINAL A* CHECKLIST

1. For every proof by contradiction: assumption in words, algebra, contradiction stated, conclusion.
2. For every skew-lines question: “do not intersect” (third equation fails) and “not parallel” (with reason).
3. For every binomial: take out a^n , bracket (bx/a) fully, simplest fractions, state $|x| < \dots$.
4. For every implicit derivative: product rule on xy terms, constants $\rightarrow 0$, dy/dx collected and factorised.
5. For every parametric integral: $y \times dx/dt$, limits in t , show the simplification for a given form.

6. For every substitution: change the integrand, du and the limits; never $u = x$.
7. For every volume: square y correctly, π outside, exact form, units in context.
8. For every differential equation: $+ c$, use the condition, rearrange to the requested form, consider $t \rightarrow \infty$.
9. For every connected-rates question: substitute the correct variable (h , r or V), not the time.
10. Leave 10 minutes: recheck signs in vector components and the exact form requested.